The 8085 Master Manual

Addendum: The Support Ecosystem

(Peripheral Controllers & Hardware Extension)

A Grassroots Engineering Guide

Moving beyond the 8085 core to understand the Intel support chips.

"A monarch is only as powerful as their ministers."

Chapter 1: The Limitations of the Core

1.1 Why Do We Need Support Chips?

Across the six volumes of the Master Manual, we built a fully functional microprocessor system. The 8085 can address memory, execute logic, and respond to interrupts.

However, as we look at the official **EE/PC/B/T/225 Syllabus**, there is one final, crucial module that we have not yet constructed. The syllabus explicitly lists:

> "Interfacing with support chips: Programmable Peripheral Interface 8255, Programmable time/counter 8253, Programmable UART 8251, Programmable Interrupt Controller 8259, DMA Controller 8257..."

THE PERFORMANCE BOTTLENECK

The 8085 is an exceptional general-purpose processor, but it is terrible at multitasking and dedicated hardware control.

If you want the 8085 to accurately measure a 1-second delay, you must trap it in a software counting loop (like we did in Volume V). While it is counting, it cannot do *anything* else. If you want it to scan a 64-key keyboard, it must constantly read input ports, wasting precious clock cycles.

To solve this, Intel designed a family of **Programmable Support Chips**. These are essentially "mini-processors" dedicated to one specific task. You wire them to the 8085's System Bus. The 8085 sends them a command (e.g., "Count to 10,000 and then interrupt me"), and then the 8085 goes back to its main program. The support chip handles the tedious hardware task independently.

This Addendum serves as the final phase (Phase 7) of your grassroots education, introducing the concept of these support ministers.

Chapter 2: The 8255 Programmable Peripheral Interface (PPI)

The most fundamental and frequently examined support chip is the **8255 PPI**.

2.1 The Philosophy of the 8255

If you want to connect 24 LEDs or 24 switches to the 8085, you would need to build complex Address Decoding logic (using NAND gates or 74LS138s) to create three separate 8-bit I/O ports.

The 8255 replaces all of that discrete logic. It provides **24 programmable I/O pins** in a single chip. It connects directly to the 8085's data bus and exposes three distinct 8-bit ports to the outside world: **Port A, Port B, and Port C**.

2.2 The Control Word Register

The true power of the 8255 is that it is *programmable*. The 8085 can send an 8-bit "Control Word" to the 8255, instructing it to configure Port A as an input (to read switches) and Port B as an output (to drive LEDs).

8255 Control Word Format (I/O Mode)

1	Mode	Mode	Port A	Port Cu	Mode	Port B	Port Cl
D7	D6	D5	D4	D3	D2	D1	D0
I/O Flag	Group A Mode		Group A I/O (1=In, 0=Out)		Group B I/O (1=In, 0=Out)		

2.3 Bit Set/Reset (BSR) Mode

The 8255 has a special feature known as BSR mode. This mode is used exclusively to set or reset individual pins on **Port C** without affecting the other pins. This is incredibly useful for generating control pulses (like strobing a specific chip on the motherboard).

To enter BSR mode, the 8085 sends a Control Word to the 8255 where the most significant bit (D7) is strictly **0**.

Chapter 3: The Expanding Ecosystem

While the 8255 handles general I/O, other specialized chips handle complex subsystems.

3.1 The 8253 Programmable Interval Timer

Instead of using software delay loops (which freeze the 8085), the system uses the 8253. It contains three independent 16-bit hardware counters. The 8085 sends a number to the 8253 and says, "Count down at 1 MHz, and send me an interrupt when you hit zero." The 8085 is then free to do other processing while the 8253 counts perfectly in the background.

3.2 The 8259 Programmable Interrupt Controller (PIC)

The 8085 only has five hardware interrupt pins. What if a motherboard has 8 or 15 different devices that need to trigger interrupts?

The 8259 PIC acts as an interrupt multiplexer. You connect up to 8 devices to the 8259, and you connect the 8259 to the single **INTR** pin of the 8085. The 8259 manages the priority of those 8 devices, and when it interrupts the 8085, it automatically places the correct `CALL` instruction onto the data bus so the 8085 knows exactly which device triggered the alarm.

3.3 The 8257 Direct Memory Access (DMA) Controller

If a hard drive wants to transfer 4 KB of data to RAM, routing every single byte through the 8085's Accumulator is incredibly slow.

The 8257 DMA controller takes absolute control of the system. It sends a signal to the 8085's **HOLD** pin. The 8085 suspends itself, puts its buses into high-impedance, and sends an **HLDA** (Hold Acknowledge) signal back. The 8257 then physically takes over the Address and Data buses, blasting data directly from the hard drive into RAM without the 8085 even knowing.

CONCLUSION OF THE GRASSROOTS JOURNEY

If you look at the 2022-2025 Past Year Questions (PYQs), you will notice that Professor Mihir Hembram focuses almost exclusively on the core 8085 architecture (Vols I - VI). Questions on the 8255 and other support chips appear very rarely, typically as short-answer theory questions (e.g., "Explain BSR mode").

You now possess a complete, ground-up understanding of the machine. You started with a blank piece of silicon and ended with a fully orchestrated, DMA-capable motherboard ecosystem. You are ready for the exam.

FINAL ADDENDUM COMPLETE